

Drawing with Code!

Make Beautiful Rangoli & Shapes with Your Digital Pet Turtle 

Core Concepts: Giving Clear Steps, Turning Corners, & Creating Magic Patterns

Tool: Python Turtle Graphics

Welcome to Your Coding Journey!

Have you ever seen the beautiful, colorful patterns drawn outside doorsteps during festive celebrations like Diwali? Those are called **Rangoli** and **Kolam**! They are made of perfectly balanced shapes like triangles, squares, and stars.

In this book, we will learn how to draw these amazing patterns using code! Instead of chalk that might slip or smudge, we are going to use a friendly, smart digital pet called **Python Turtle**. Our Turtle follows your directions perfectly every single time to make beautiful, glowing art!

Module Roadmap: What We Will Learn

Level	Your Exciting Mission	Coding Superpower
01	Meet Your Digital Turtle Pet	Art Meets Coding
02	Code Instructions as Footsteps	Doing Things Step-by-Step
03	Steering Your Turtle (The Bike Handlebar Rule)	Turning Left and Right
04	Building Magic Symmetrical Mandalas	The 360° Circle Rule
05	Live Lab 1: Walking Forward!	<code>turtle.forward()</code>
06	Live Lab 2: Spinning in Place!	<code>turtle.right()</code>
07	The Square Box Architect Challenge	Building Your First Shape
08	Code Doctor: Fixing Broken Code	Finding and Fixing Bugs

Module 01: Meet Your Turtle Pet!

Ananya and Kabir's Story:

Ananya was drawing a beautiful chalk design on the ground for a big family celebration. Suddenly—oops! The chalk slipped and smudged a perfect line. Kabir smiled and had a great idea: "Let's use Python Turtle! It's a digital pet that paints perfect art with pixel precision every single time!"

When we code, a small slip doesn't ruin your canvas! If you make a mistake, you just change a word of text, press run, and watch the world rebuild itself instantly.

Module 02: Code Instructions as Footsteps

Computers read instructions exactly like a dancer follows choreography steps—**one single clean step at a time, from top to bottom**. If you mix up the order of the dance steps, the whole performance changes!

The Dance Analogy: Just like dance beats guide you step-by-step across the room, our Python Turtle pet follows your numbered moves in the exact order you write them!

 Dance Routine Steps	 Turtle Code Steps
1. Take a big step forward.	1. <code>turtle.forward(100)</code>
2. Turn your shoulders to the right.	2. <code>turtle.right(90)</code>
3. Take another big step forward.	3. <code>turtle.forward(100)</code>

Module 03: Steering Your Turtle

To guide your turtle smoothly, you need to master turns. Turning your turtle's nose is exactly like **steering a bicycle's handlebar** or spinning a spinning top (Lattu)!

When you tell the turtle to turn, it stays on the exact same spot on the screen, but its nose points to a brand-new direction.

```
import turtle
turtle.right(90) # This spins your turtle clockwise to face South!
```

Module 04: Creating Magic Mandalas

True art magic happens when you combine simple shapes into giant, beautiful masterpieces! By drawing a simple square and spinning it around multiple times, you can create a glowing geometric flower mandala.

The Magic 360° Circle Rule: To make a shape or a pattern close completely without any messy gaps, all your turns must add up to exactly **360 degrees** (a full circle)!

Shape Blueprint Guide:

- **Classic Square:** Turn 90 degrees, repeat 4 times!
- **Equilateral Triangle:** Turn 120 degrees, repeat 3 times!
- **Symmetrical Hexagon:** Turn 60 degrees, repeat 6 times!

Module 05: Live Lab 01 - Walking Forward

Let's try typing code in our sandbox! Type this exact code into your editor:

```
import turtle
turtle.speed(3)      # Sets a nice, friendly speed
turtle.forward(100)  # Tells our turtle to take 100 small steps forward
```

What happens? Your digital turtle marches 100 pixels forward, leaving a sharp, bright line behind it like a glowing marker!

 **Brain Teaser:** If you type `turtle.forward(50)` right after, does the turtle jump back to the center or keep moving from where it stopped? (Try it and see!)

Module 06: Live Lab 02 - Spinning in Place

Now, let's learn to turn without walking. Type this command:

```
turtle.right(90) # Spins the turtle like a compass needle!
```

Notice that the turtle doesn't draw a line when turning—it just pivots its body in place. Let's combine them to make a sharp **L-shape**:

```
turtle.forward(100)  
turtle.right(90)
```

Module 07: The Square Box Architect Challenge

It's time to build your very first complete shape! You are now a master structural architect. Let's write a program to build a perfect square box:

```
import turtle

# Side 1
turtle.forward(100)
turtle.right(90)

# Side 2
turtle.forward(100)
turtle.right(90)

# Side 3
turtle.forward(100)
turtle.right(90)

# Side 4
turtle.forward(100)
turtle.right(90)
```

Architect Checklist:

- Did the forward and right pattern repeat exactly 4 times? Yes!
- Did the turtle return home to the start, facing the original direction? Yes!

Module 08: Code Doctor Sandbox

Sometimes code has mistakes called **Bugs**. When a bug happens, the computer gets confused and prints an error message. Let's learn how to fix them!

Broken Code 	The Problem	How to Fix It Tool 
<code>turtleforward(100)</code>	Missing Dot	Add a dot: <code>turtle.forward(100)</code>
<code>turtle.right(90</code>	Missing Bracket	Add closing bracket: <code>turtle.right(90)</code>
<code>Turtle.forward(50)</code>	Capital 'T'	Use lowercase 't': <code>turtle.forward(50)</code>



CODE DOCTOR CHALLENGE: Can you fix this broken line so it doesn't crash?

`turtel.rihgt[90)`



Quick Quiz Checkpoint

1. Which command turns your turtle's nose without moving its position?

2. What shape do you get if you move forward 100 steps and turn 90 degrees exactly 4 times?



Home Quest: Pattern Hunter Challenge

Your mission tonight is to look around your home and find real-world square patterns (like floor tiles, a window pane, a chessboard, or a picture frame). Write down step-by-step instructions in plain English to recreate it below!

1. Shape I found:

2. My Step-by-Step Directions: